

Lab 3, Act 3 – Answer Key

From counts to a continuous measurement

NRES 746 – Instructor Answer Key (not for student distribution)

This key follows the same section structure as the student handout. Questions with a single correct numeric answer are answered directly; open-ended items (interpretation, discussion) are given sample answers/discussion points rather than a fixed “correct” response.

Part 1: Reading a density curve

1a. The shaded region should span from mass = 4.5 to mass = 5.5 under the curve – the area under that stretch of curve, not a slice of the axis or a vertical strip at a single point.

1b. Integral notation.

$$P(4.5 < \text{mass} < 5.5) = \int_{4.5}^{5.5} f(x) dx$$

1c. $P(\text{mass} = 4.5) = 0$ exactly. A single point has zero *width*, so the “area” under the curve over just that point is zero, regardless of how tall the curve is there – probability for a continuous random variable only accumulates over an interval (a range with nonzero width).

1d. The height of the curve is a **probability density**, not a probability. It represents how concentrated probability is near that value of x – probability *per unit* of x – which is why it can be compared across regions (taller = more probability mass packed in nearby) but only becomes an actual probability once multiplied by (or integrated over) some width.

Part 2: Reading a CDF

2a. Reading $F(5.5) \approx 0.95$ off the plot and subtracting gives $P(\text{mass} > 5.5) \approx 0.05$ ($\approx 5\%$). Any answer in the neighborhood of 0.03-0.08 from graph-reading alone is reasonable – precision comes in Part 3.

Part 3: Standardize and look it up

3a.

```
z <- (5.5 - mu) / sigma
z
```

```
## [1] 1.666667
```

$z \approx 1.67$.

3b.

```
1 - pnorm(1.67)           # table-precision version, Phi(1.67) from Table 1
```

```
## [1] 0.04745968
```

```
1 - pnorm(5.5, mu, sigma) # exact, for comparison
```

```
## [1] 0.04779035
```

Table gives $\Phi(1.67) \approx 0.9525$, so $P(\text{mass} > 5.5) \approx 1 - 0.9525 = 0.0475 \approx 4.75\%$ (exact value via `pnorm` is 0.0478 – the small gap is just table rounding to two decimals of z). This should be close to the 2a graph-reading estimate.

3c.

```
pnorm(1.28)           # closest table entry to 0.90
```

```
## [1] 0.8997274
```

```
mu + 1.28 * sigma     # convert back to a body mass
```

```
## [1] 5.268
```

```
qnorm(0.9, mu, sigma) # exact, for comparison
```

```
## [1] 5.268931
```

The table's closest entry to 0.90 is $\Phi(1.28) = 0.8997$, giving $z \approx 1.28$ and a cutoff of $4.5 + 1.28(0.6) \approx 5.27$ kg (exact value via `qnorm` is also 5.27 kg). **Watch for** students who report $z = 1.28$ itself as the answer and forget to convert back into a body mass – the table gives z , not x .

Group discussion and revision

Discussion prompt answer: $z = (x - \mu)/\sigma$ is valid arithmetic for *any* distribution with a defined mean and standard deviation – that part doesn't depend on normality. What's special here is that for a variable that really is normal, standardizing produces *exactly* the standard normal curve every time, regardless of μ and σ – that's the one curve this table encodes. If you standardized a variable that wasn't normal to begin with (e.g. Act 2's binomial detection counts, which are discrete and right-skewed for $p = 0.3$), you'd still get a rescaled variable with mean 0 and variance 1, but its actual *shape* would still be discrete and skewed, not bell-shaped – so looking up $\Phi(z)$ in this table would give the wrong probability for it. Two common wrong answers worth watching for and gently correcting:

- **“It works because of symmetry”** – necessary but not sufficient. A uniform distribution is symmetric and standardizes to mean 0/variance 1 too, but its standardized shape is flat, not bell-shaped, so this table would still give the wrong answer for it.
- **“It works because we're not near a hard boundary (like 0)”** – this is a real and relevant consideration, but it's about whether *assuming* a normal model is reasonable for this particular data in the first place (body mass can't be negative), not about whether the standardizing arithmetic is valid once you've already decided to treat something as normal.